



WS & SOA (IT644)

CodeTrail

Name	Student ID
Monar	202512019
Rishika Shah	202512041
Meet Sheth	202512057
Ruchita Patadiya	202512070
Shaikh Maryam	202512074
Mahi Kansara	202512078
Anistina Dsouza	202512081
Thakkar Manushri	202512122

CodeTrail

Build Together. Prove Every Contribution.

CodeTrail is a **web-based platform** that allows various users to write, edit, and carry out code at the same time in a **common workspace**. Because it makes use of real-time communication technologies, it instantly synchronises the changes to the code across all the users who are connected, thus giving a smooth experience when programming together.

In addition to enabling real-time collaboration, CodeTrail automatically keeps a **record of each participant's coding activities** during a session. The record of each person's contributions thus provides a clear account of individual participation, which promotes accountability during collaborative development processes.

The platform is aimed at students who are working on academic projects, at those taking part in hackathons, at freelance developers, at remote software teams, and at other kinds of collaborative programming situations. CodeTrail offers an efficient and transparent environment for **collaborative software development** by bringing together real-time collaborative coding, code execution directly in the browser, and automatic tracking of contributions all within a single web application.

Application Type

CodeTrail is a **web-based application** which offers a single platform for both administrators and end users (developers/collaborators) via a conventional web browser.

The reason for choosing a web application rather than a mobile or hybrid one is that the main feature of CodeTrail – **real-time collaborative code editing** – needs a feature-packed development environment with a full-size code editor, a file

explorer, an output console, live cursor visualization, and a number of different interface panels all working at the same time.

The application features a **responsive and interactive user interface** which is accessible in all modern web browsers without users having to install any additional software by using **React.js** on the frontend. This method also makes deployment, maintenance and updates easier since it involves keeping a single codebase and at the same time ensures compatibility across various operating systems and devices thanks to browser support.

Motivation / Problem Statement

Because of the growing use of **remote work, online education, hackathons, and the development of software on a distributed basis**, real-time collaborative programming has now become an essential element of modern software development. **Developers, students, and freelancers** frequently work together on the same codebase while being at different locations. Even though the current collaborative code editors do provide real-time code synchronization, they mainly aim at facilitating collaboration and give only limited insight into the individual contributions made during a coding session.

This leads to a number of practical difficulties. In academic group projects teachers might have trouble assessing each student's real contribution. In the same way, freelancers, remote teams and hackathon participants usually do not have a **reliable and transparent** method of showing individual work or proving their contributions when developing together.

Time-tracking programmes do record how long work takes, but they depend on information being entered by the user and therefore do not offer any proof of actual coding activity. In a similar way, version control systems keep records of changes yet they do not produce structured, session-based reports which clearly show each person's level of involvement.

In order to overcome these limitations, CodeTrail combines **real-time collaborative coding** with an **automated system for tracking contributions**. The platform keeps a record of **important coding activities** in order to produce a clear account of each person's contributions. By bringing together collaborative editing, browser-based code execution, and contribution tracking in a single web application, CodeTrail seeks to enhance collaboration, transparency, and accountability for its users.

Target Users and Use Cases

Target Users

1. Students and Academic Teachers

The intended users are students, professors, and teaching assistants.

CodeTrail allows students to work together on their programming assignments, academic projects, and hackathons at the same time as enabling teachers to look at each student's individual contributions.

2. Freelance Developers and Clients

The intended users are freelancers, software agencies, and clients.

The platform supports collaborative development and at the same time keeps a clear record of the coding activities and the individual contributions.

3. Remote Development Teams

The intended users are software developers, engineering teams, and technical interviewers.

CodeTrail enables remote collaboration, pair programming, and coding interviews by means of a shared coding workspace.

4. Platform Administrators

Target Users: System administrators.

Administrators are in charge of users, keep an eye on collaboration sessions, and supervise the platform's performance.

Primary Use Cases

Use Case	Primary Actor	Description
UC-01: Register and Authenticate User	User	Register an account, log in securely, and access the application using authenticated sessions.
UC-02: Create and Manage Workspace	Workspace Owner	Create a collaborative workspace, invite collaborators, configure workspace settings, and manage participants.
UC-03: Collaborate on Code in Real Time	Collaborator	Join a shared workspace and edit source code simultaneously with other users using real-time synchronization.
UC-04: Execute Source Code	Collaborator	Compile and execute code in different programming languages and view execution results within the application.
UC-05: Track User Contributions	System	Automatically record coding activities and maintain a secure, timestamped contribution history during collaboration sessions.

Scope and High Level Features

Scope of the Project

The aim of CodeTrail is to create a **web-based collaborative coding platform** on which several users can write, edit, carry out, and manage source code in real time, with the system automatically keeping record of each person's contributions. The platform is intended for use by educational institutions, independent developers, remote teams, and participants in hackathons who need **efficient collaboration and clear records of individual contributions**.

The system offers a **secure environment** that works through a browser, enabling users to set up coding sessions, invite people to join, and collaborate together without having to install any software locally. All coding activities, such as editing, running code, and taking part in a session, are automatically logged so that there is a clear record of everyone's contributions.

CodeTrail also produces contribution reports in order to aid with performance evaluations, ensure transparency in collaboration, and promote accountability.

The project only includes browser-based collaborative programming and the tracking of contributions and has no intention of replacing full-featured IDEs, version control systems, or enterprise project management tools.

High Level Features of the Project

1. User Authentication & Authorization

This module will provide secure user authentication, authorization, and account management to ensure that only authenticated users can access collaborative workspaces and platform features.

2. Workspace & Collaboration Management

This module will enable users to create, join, and manage collaborative coding workspaces while controlling participant access and session management.

3. Real-Time Collaborative Code Editor

This module will aim to provide the users a shared coding environment where multiple users can edit source code simultaneously with real-time synchronization and collaboration features.

4. Code Execution Engine

This module will make sure that users can compile and execute source code directly within the browser using a cloud-based execution service.

5. Activity Logging & Proof-of-Work Tracking

This module will automatically record the coding activities and maintain a secure, timestamped, and tamper-evident history of individual contributions throughout each collaboration session.

6. Contribution Analytics & Report Generation

This module will analyze user contributions and generate detailed reports that provide insights into participation, coding activity, and overall collaboration.

7. Dashboard and Project History

This module will provide all the users with a centralized dashboard to access recent workspaces, contribution history, project summaries, and previously generated reports.

8. Workspace Settings & Notifications

This module enables users to configure workspace preferences, manage collaboration settings, and receive real-time notifications about important workspace activities, improving communication and the overall collaboration experience.

The Technology Stack

CodeTrail makes use of the MERN Stack together with various supporting technologies which enable real-time collaboration, secure authentication, code execution, activity tracking, and the generation of reports.

Frontend

React.js is used for creating an interactive and responsive user interface.

Vite is a development and build tool designed to speed up development.

Browser-based code editor with syntax highlighting and support for multiple languages

Tailwind CSS provides responsive and modern styling for user interfaces.

Backend

Node.js is a runtime environment used on the server side for handling application logic and real-time requests.

Express.js is a framework used for the creation of RESTful APIs and backend services.

Database

MongoDB contains the users, workspaces, files, activity logs, and reports.

Mongoose is an Object Data Modeling (ODM) library for MongoDB.

Real-Time Collaboration

Socket.IO allows for the synchronization of code in real time, the updating of cursors live, and collaborative editing.

Authentication & Security

JWT is used for secure user authentication and for session management.

bcrypt is a method for hashing and encrypting passwords.

CORS allows for secure communication between the frontend and the backend.

Input Validation checks user input in order to enhance the security of the application.

Contribution Verification

The **SHA-256 hashing method** produces tamper-evident activity logs by securely associating recorded events.

Report Generation

jsPDF is capable of producing downloadable PDF reports for contributions and sessions.

Development Tools

Visual Studio Code is a source code editor.

Version control and repository management using **Git and GitHub**.

Postman is involved in API development and testing.

MongoDB Atlas is a MongoDB database that is hosted in the cloud.

npm handles packages and dependencies.

Roles and Responsibilities

Module-based Work Distribution

In order to achieve a **fair sharing of the workload** and to **promote effective cooperation**, the project adopts a development method based on modules instead of the conventional division between frontend and backend. The development team is divided into two teams, each consisting of four members, and the project modules are allocated between the teams.

The responsibility for **fully implementing the modules** that have been assigned lies with each team, covering frontend development, the development of backend services, database integration, testing, debugging, and documentation. Because each team has end-to-end control of its modules, development can proceed in parallel, the integration of the modules is improved, and each team member is able to gain practical experience in the whole software development lifecycle.

The module allocation is as follows:

Team	Assigned Modules	Responsibilities
Team A Manushri Thakkar Ruchita Patadiya Mahi Kansara Monar	Module 1: User Authentication & Authorization Module 3: Real-Time Collaborative Code Editor Module 5: Activity Logging & Proof-of-Work Tracking Module 7: Dashboard & Project History	Design the assigned modules collaboratively , including work on the frontend, backend, database integration, and API development , and carry out the testing, integration, and documentation.
Team B Maryam Shaikh Anistina Dsouza Rishika Shah Meet Sheth	Module 2: Workspace & Collaboration Management Module 4: Code Execution Engine Module 6: Contribution Analytics & Report Generation Module 8: Workspace Settings & Notifications	Take part in code reviews, debugging, and the integration of modules.

Team Collaboration

Even though the project modules are divided between the teams, all the members will carry out active collaboration during the entire project lifecycle. The two teams will take part in the requirements analysis, the design discussions, the code reviews, the module integration, testing, debugging, deployment, documentation, and the final project presentation in order to ensure the successful delivery of the complete application.

Timeline and Milestones

Month	Week	Milestone	Development Activities	Expected Deliverable
July	Week 1	Team Formation & Project Ideation	Team formation, literature survey, project discussion, technology evaluation, and project idea finalization.	Project group formation & Project idea submission
August	Week 2	Project Proposal	Define project scope, objectives, application type, technology stack, module breakdown, roles and responsibilities, and project timeline.	Project Proposal Document
August	Week 3	Project Planning	Prepare the Work Breakdown Structure (WBS), estimate person-hours, assign modules, create the GitHub repository, and configure project management tools (Trello/Jira).	Project Plan
August	Week 4	Requirements Engineering	Gather functional and non-functional requirements, prepare user stories, use case diagrams, interface requirements, wireframes, and acceptance criteria.	Business Requirements Document (BRD)
August	Week 5	System Design	Design the software architecture, database schema, REST API contracts, UML diagrams, and finalize UI/UX mockups.	System Design Document

September	Week 6	Module 1 – User Authentication & Authorization	Implement secure registration, login, JWT authentication, user profile management, and role-based access control.	Authentication module completed
September	Week 7	Module 2 – Workspace & Collaboration Management	Develop collaborative workspaces, participant management, invitation system, and session management.	Workspace management module completed
September	Week 8	Module 3 – Real-Time Collaborative Code Editor	Integrate Monaco Editor, implement real-time synchronization using Socket.IO, and develop collaborative editing features.	Collaborative editor completed
September	Week 9	Module 4 – Code Execution Engine	Integrate the Piston API, implement multi-language support, execution console, and execution history.	Code execution module completed
October	Week 10	Module 5 – Activity Logging & Proof-of-Work Tracking	Develop activity logging, contribution tracking, timestamp recording, and SHA-256 hash-based verification.	Contribution tracking module completed
October	Week 11	Mid-Semester Demonstration	Integrate completed modules, perform preliminary testing, resolve issues, and demonstrate the working prototype.	Mid-Semester Demonstration
October	Week 12	Module 6 – Contribution Analytics & Report Generation	Develop analytics dashboard, contribution summaries, session reports, and PDF report generation.	Analytics & reporting module completed

November	Week 13	Module 7 – Dashboard & Project History	Implement personalized dashboards, workspace history, activity timeline, and project summaries.	Dashboard module completed
November	Week 14	Module 8 – Workspace Settings & Notifications	Develop workspace configuration, collaboration settings, notification system, and integrate all functional modules.	Workspace settings & notifications completed
November	Week 15	System Integration & Testing	Perform module integration, unit testing, integration testing, UI testing, database population, bug fixing, and validation.	Test Plan & Test Cases
November	Week 16	Final Application & Deployment	Finalize the application, deploy to a public platform, prepare API documentation, README, user manual, and deployment notes.	Final Application, Deployment & Hosting
November	Week 17	Final Documentation & Project Presentation	Prepare the final project report, presentation slides, project poster, video walkthrough, peer review, and demonstrate the completed application.	Final Report, Final Presentation & Demo, Peer Review & Contribution Report